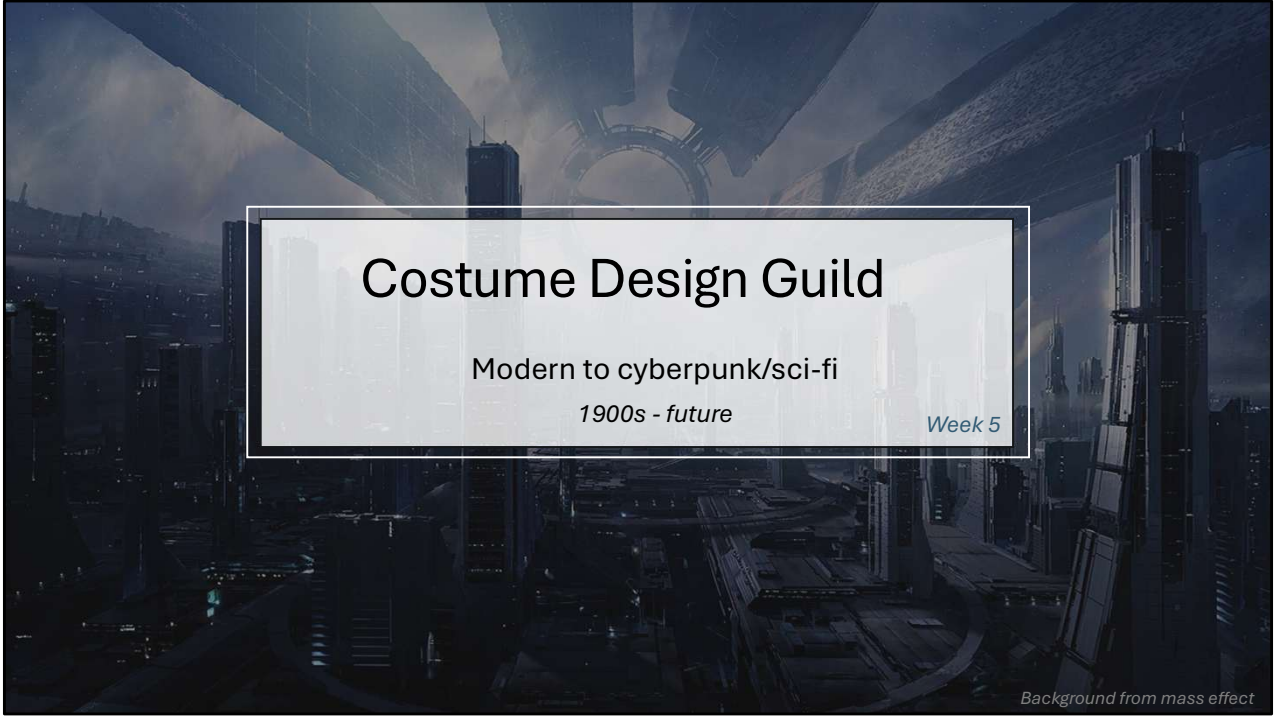




Costume Design Guild

Modern to cyberpunk/sci-fi
1900s - future

Week 5

Background from mass effect

My master research

- Fashion designer > master game technologies
- Running these guilds as part of my master research
- Needs to be recorded for documentation
- Your input is my data
- Helps me graduate ☺
- In turn: Ask me for help, feedback etc on your costume work!



To introduce myself quickly, I am Imara, my previous study was fashion design and I am now doing my master in game technologies.

I've started running these guilds as part of my master research, and it's about figuring out which costume design principles are important for video game design,

and I will find this out by observing how you guys interact with and react to the guild content I have prepared for you guys

Your input, thoughts, reflections, work you show me if all part of the data I'm gathering. Your input will help me graduate

And in turn I am always available if you need help, feedback or anything else.

Acknowledgements

- Most of the periods in this session are speculative
- I will show you stereotypes
- Give you ideas to think further with
- Not facts, you can invent anything



I want to be transparent about a few things in this topic

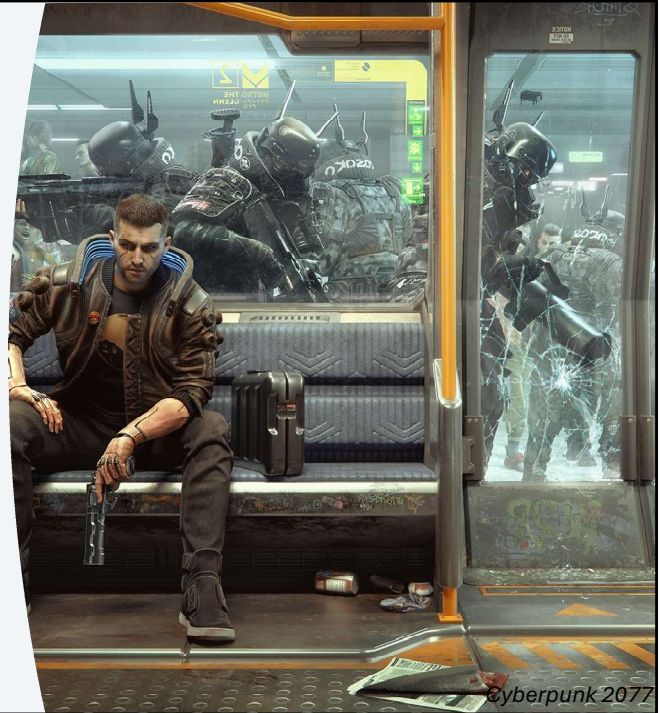
Most of the periods in this session are speculative, this means it doesn't exist yet

I will show you stereotypes and give you ideas to think further with

These are not facts and you can invent anything beyond the current limits.

Agenda

1. Period & game examples
2. Silhouette & key garments
3. Social hierarchy
4. Materials, textiles & dyes
5. Construction & details
6. Special topics
7. Discussion & reflection



The agenda of today is the same as last guilds, covering the whole range of costume principles within the time period of modern to future

Period & game examples

Section 1

Starfield

I'll start with the period and game examples

Term definition:



contemporary

/kənˈtɛmp(ə)r(ə)rɪ/

adjective

1. **living or occurring at the same time.**

"the event was recorded by a contemporary historian"

2. **belonging to or occurring in the present.**

"the tension and complexities of our contemporary society"

Similar:

modern

present-day

present

current

present-time



noun

a person or thing living or existing at the same time as another.

"he was a contemporary of Darwin"

Similar:

peer

fellow

compeer

coeval

Lets start with an important word you need to know before diving into the period definition

Contemporary in this case generally means modern times, so the present day. So if I talk about contemporary fashion, I talk about the fashion we are wearing right now.

Period explanation

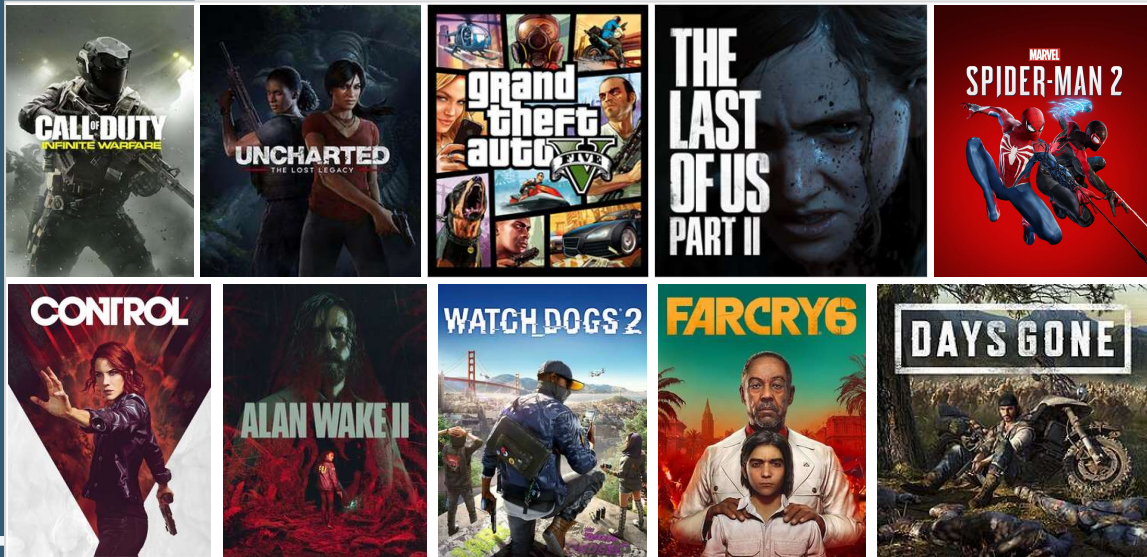
- **Second largest category (~20-25% of AAA games)**
- No historical reference = pure design problem-solving
- Asset bashing skills essential
- Contemporary relatability



Okay so This period is the second largest category used to inspire triple A games There are no references yet for the future so it's a lot of design problem solving Asset bashing skills are essential as this period can combine a bunch of different things to create new designs

And this period feels most relevant to the modern-day audience, which can also be called contemporary relatability. Which is the ability of a character, in this case, to feel relevant, understandable and it bridges the gap between the past and present

Game examples

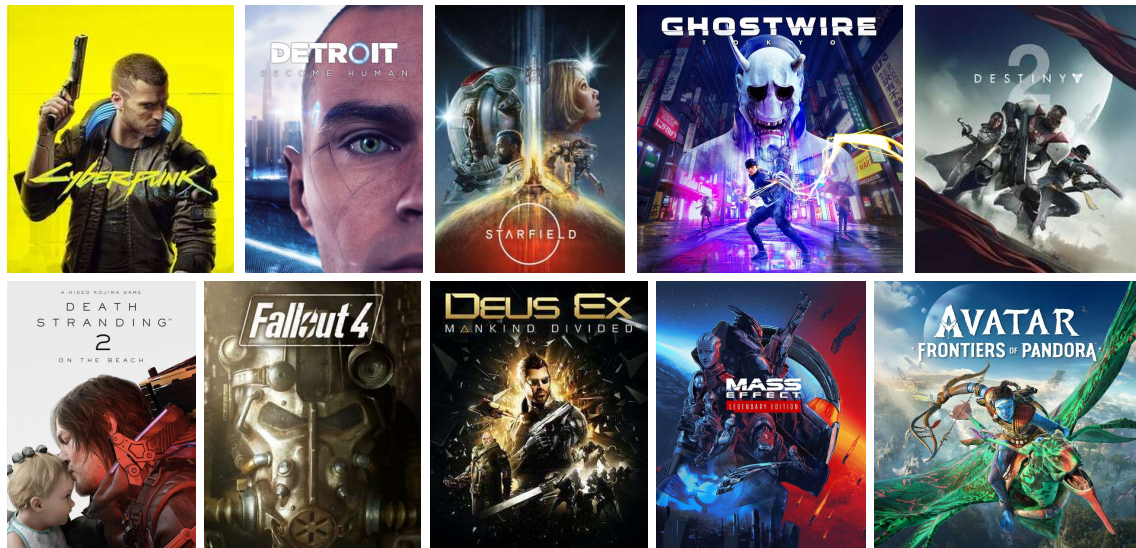


Modern 1920-2050

Here is a bunch of games set in the modern time.

My personal favourite is control, the clothing in the game are actually made using scanning or marvelous designer simulations and the process can be found online so that makes the game even cooler to me

Game examples

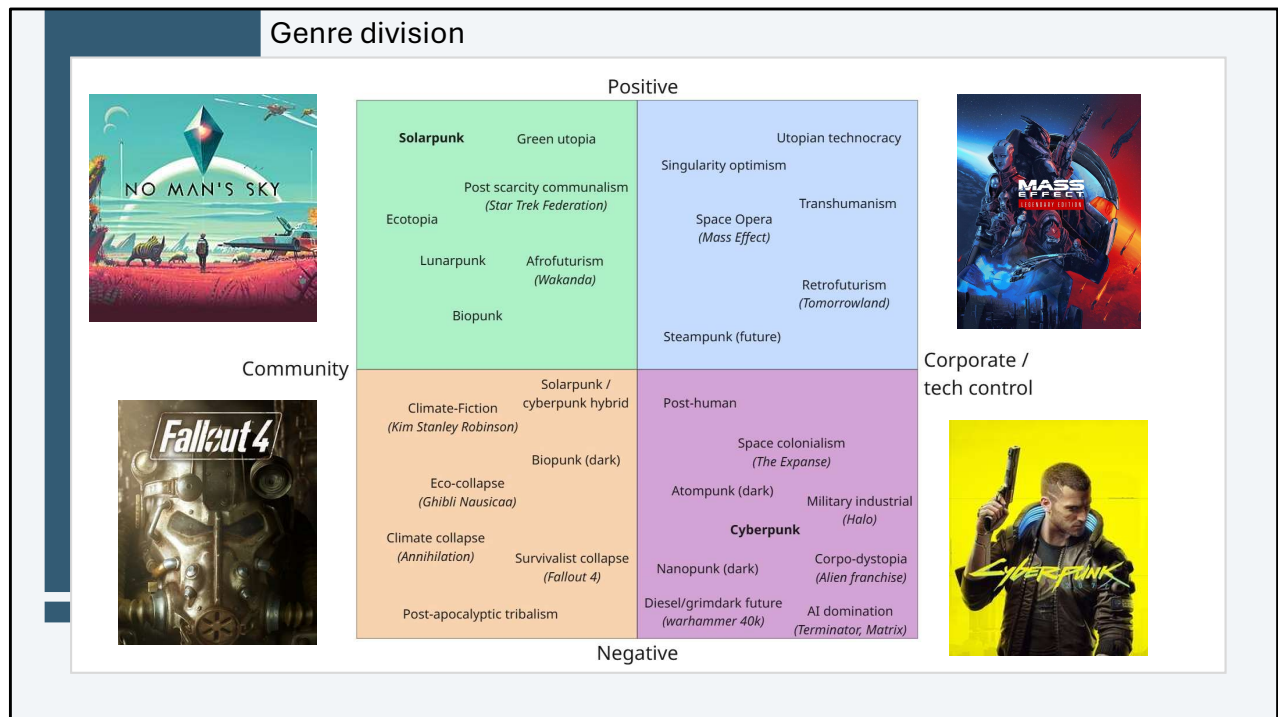


Future 2050+

And here are some of the popular games set in the future

You probably have played a few titles from both time periods so you will recognise examples I will show later on

My personal favourite here is death stranding, also a lot of marvelous used for it



I've created a quick quadrant graph to show you different genres of the future. You can use this as keywords to google or find more references to dive deeper into a genre

I've added popular games so you can visualise what each quadrant might look like which might help you decide what aesthetic you want to go for if you are designing something futuristic.

I'm uploading all powerpoints to teams so you can look at this deeper after this guild session if you want

Key take-aways

- When there's no historical reference, how do you design believable clothing?
- This period teaches you to COMBINE elements from different eras (asset bashing)
- And think about how materials and technology shape what people wear.



The key takeaways from this time period is helping you think about how to design believable clothing when theres no historical reference yet
This period will allow you to combine elements and think about how materials and technology shape what people wear

Quick question

- What's your favourite game from this era and why?



Silhouette & key garments

Section 2

COD Ghost outfit

The next topic is silhouette and key garments

Defining silhouette

MODERN CONTEMPORARY:

- Fitted, practical, recognizable
- Jeans, t-shirts, jackets = baseline
- Layering for visual interest



TACTICAL/MILITARY:

- Modular, pouches, straps
- Body armour plates (chest, shoulders)
- Utility-focused silhouette



Contemporary clothing can be defined by fitted, practical and recognisable silhouettes

Most seen is the jeans, tshirt and jackets silhouette where layers create visual interest

Then for the tactical and military silhouette, the modular gear is important. The silhouette is created by utility pieces like the body armour, pouches, attachments etcetera

Defining silhouette

POST-APOCALYPTIC:

- Layers of scavenged materials
- Makeshift armour (tire rubber, road signs, sports gear)
- Weathered, patched, practical



The post apocalyptic silhouette can be defined by scavenged pieces and makeshift outfits.

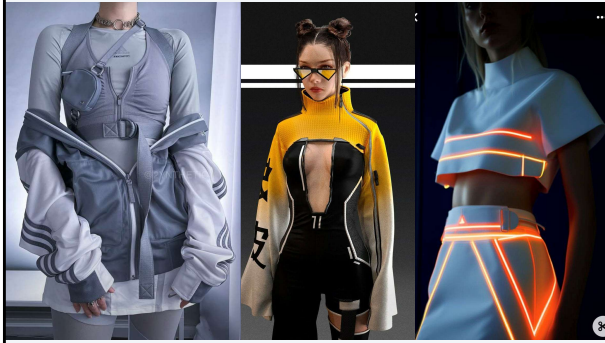
In an apocalypse, you can't really choose your outfits like we do nowadays, so you have to make do with what you can find.

And as you can't just get new things, most items will be weathered or patched and chosen for practicality.

Defining silhouette

CYBERPUNK:

- Asymmetrical - One sleeve, one leg different
- Layered (jacket over hoodie over shirt)
- Tech integration (glowing elements, ports, wires)
- High collar + low rise common combo



SCI-FI FUTURE:

- Sleek bodysuits (form-fitting base)
- Modular armor plates
- Minimal fabric, maximum function
- Glowing accents/tech details



The cyberpunk silhouette is stereotypically defined by asymmetrical clothing or details

A bunch of layers, including tech integrations like lights or wires and the combination of longer and shorter garments is often seen

Then the sci fi future stereotypical silhouette can look something like this, often with a bodysuit as base layer, modular armour plates on top.

This period still continued the cyberpunk detail of lights or tech but even more advanced as sci fi is about the far future

Key garments

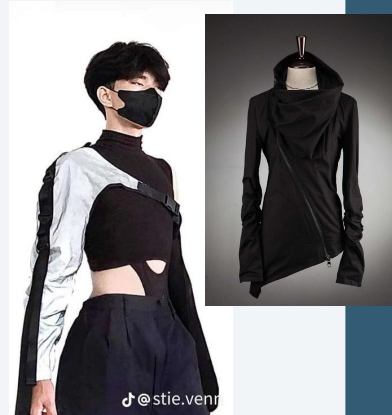
- **Modular armor system**
 - Chest plate, shoulder pads, knee guards (mix and match)



- **Tactical vest**
 - Foundation for modern military



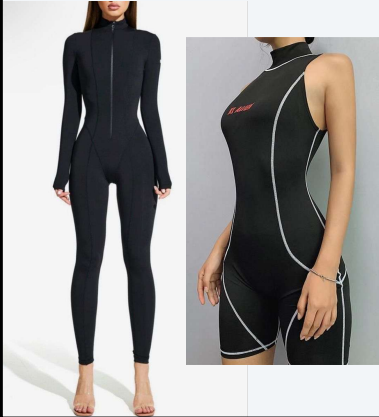
- **Asymmetrical jacket**
 - Cyberpunk staple



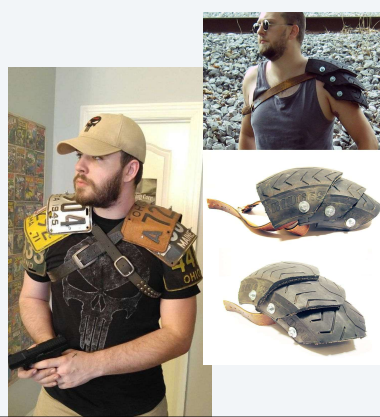
Some key garments that range this whole period of modern and future are the modular armour system, tactical vests and asymmetrical jackets

Key garments

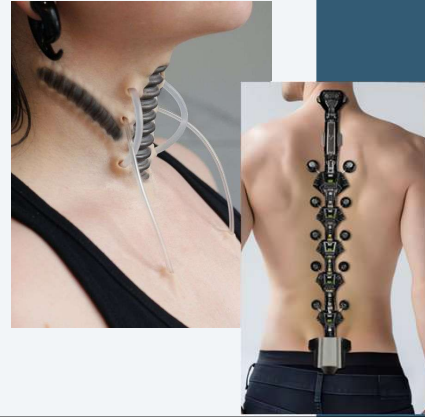
- **Bodysuit base layer**
 - Sci-fi foundation



- **Scavenged armour pieces**
 - Post-apocalyptic creativity



- **Tech integration points**
 - Where cybernetics attach

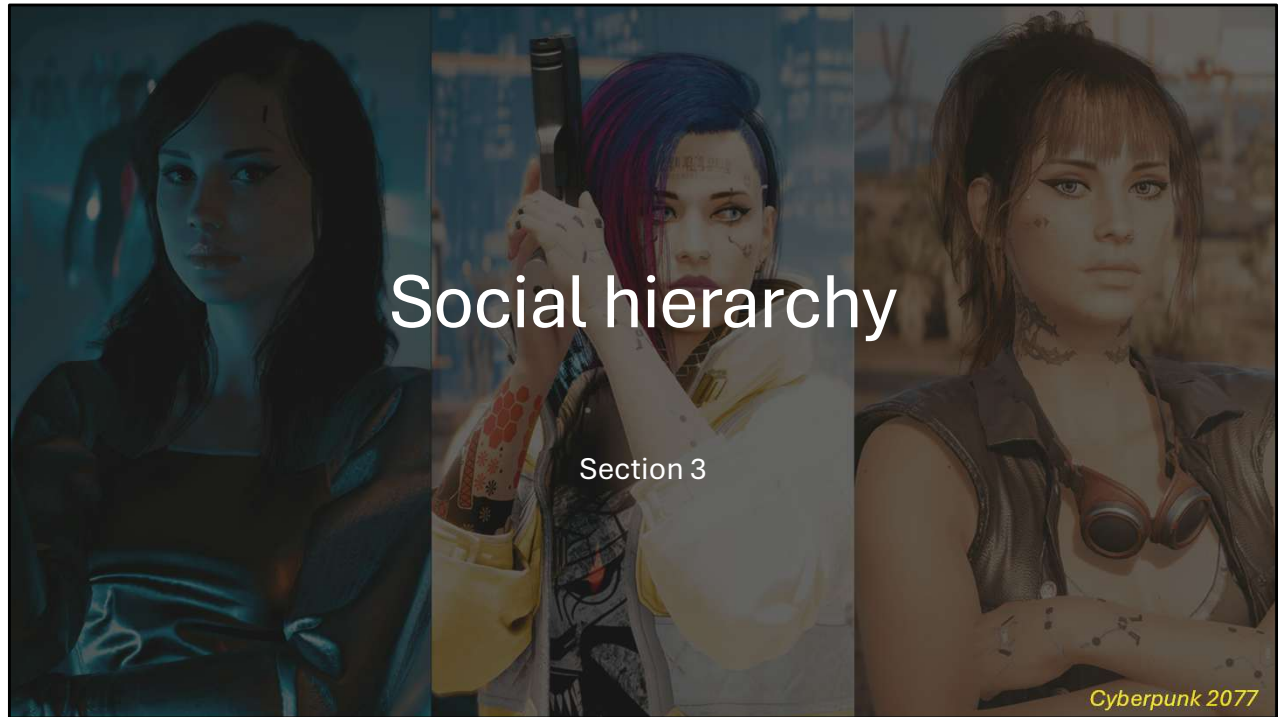


The bodysuit base layers, scavenged armour and tech integration. Whether in the body or in clothing.

Question for you

- Anything that sparks ideas for previous, current or future projects?





The next topic is social hierarchy

Visual markers of status

MODERN CONTEMPORARY:

WEALTHY → MIDDLE CLASS → WORKING CLASS → POOR

- **Wealthy:** Timeless, designer brands, perfect fit, condition, accessories (watch, bag)
- **Middle:** Mass-market brands, visible logo's, decent condition
- **Poor:** Worn, ill-fitting, outdated styles



The visual markers of status in the modern contemporary day can be seen in a few things

Wealthy people dress in designer brands with minimal logo's, perfect fit and condition with expensive yet great quality accessoires

Middle class often dress in mass market brands like zara, h&m, or higher end like stone island or Patagonia. This class often wears large visible logos, decent quality but often fast fashion so it is out of fashion quick or breaks down easily. And the lower class often wears worn or ill fitting garments with outdated styles, often from thrift stores

Visual markers of status POST-APOCALYPTIC:

WARLORDS/LEADERS → SCAVENGERS → SURVIVORS → DESPERATE

- **Leader:** BEST scavenged materials, functional armour, surplus resources (can afford decoration)
- **Scavenger:** Practical, layered, patched, worn but functional
- **Desperate:** Barely holding together, rags, makeshift

Status through RESOURCES:

- Clean water = can clean clothes (visual status)
- Surplus materials = more layers, better condition
- Weapons/armour quality = access to better scavenging



< Furiosa
End of movie Mad Max



^ Scavenger



Desperate >

The visual markers of the apocalypse kinda depend on what apocalypse is happening. Think of aliens, zombies, illnesses.

But most commonly seen the leader of a group is often dressed in the best clothing that was found, or functional armour that also looks good

Scavengers dress practically, layered garments which are often patched and worn but still functional

And the desperate or outcasts stereotypically wear barely holding it together outfits of rags and makeshift armour if any.

In the apocalypse your status can be shown through resources. Like clean water gives clean clothes, surplus materials shows in more layers or better condition And if you have access to craftsmanships like smiths or leatherworkers, the weapons and armour quality will go up significantly

Visual markers of status CORPORATE CYBERPUNK:

Arasaka executive >
Cyberpunk 2077



EXECUTIVES → CORPORATE WORKERS → STREET LEVEL → OUTCASTS

- **Executive:** Sleek, tailored, integrated HIGH-END tech (seamless)
- **Corporate Worker:** Uniform quality (company issued), basic tech
- **Street Level:** Mismatched, some or DIY tech integration, scavenged
- **Nomad/outcast:** Broken tech, patched, makeshift

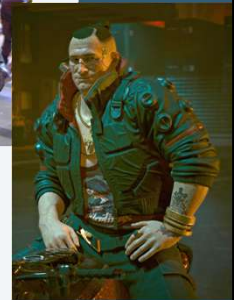
Status through AUGMENTATION:

- **High-end cyberware:** Looks organic, seamless, branded (Kiroshi optics)
- **Mid-tier:** Functional but visible, some branding
- **Low-tier:** Obviously mechanical, clunky, exposed, malfunctioning



< Corporate worker
Meth - Altered Carbon

Street level (Jackie)>
Cyberpunk 2077



The visual markers of corporate cyberpunk are still grounded in quality but have tech added to it. The more seamless, the higher in the hierarchy you are.

Here, status shows through augmentation

cyberware: Looks organic, seamless, and often branded

Mid-tier: is Functional but visible, with some branding

Low-tier: looks Obviously mechanical, clunky, exposed, and often malfunctioning

Visual markers of status SCI-FI FUTURE:

ELITE → CITIZENS → WORKERS → UNDERCLASS

- **Elite:** Latest tech, pristine, custom, highest-quality materials (may look minimalist but EXPENSIVE)
- **Citizens:** Standard-issue, functional, clean
- **Workers:** Practical, worn, lower-tech
- **Underclass:** Broken tech, patched, outdated

Status through TECHNOLOGY INTEGRATION:

- Holographic displays
- Material technology (smart fabrics, self-cleaning)

Dune elites >



*^ Citizens / workers
Zion in The Matrix Reloaded*



*Underclass >
Dredd (2012)*

The visual markers of sci fi future, which is even further in the future than cyberpunk, often goes the other way.

Tech isn't often seen on the body but in gadgets, phones or vehicles.

Here again, materials and construction or tailoring show your place in hierarchy. The status is shown through technological integration, or lack of.

Game design application

- **Cyberpunk Corporate Boss?** Over the top suit, seamless high-end cyberware, EXPENSIVE
- **Street Merc NPC?** Asymmetric jacket, visible DIY cybernetics, layered, worn
- **Post-Apoc Raider Leader?** Best scavenged armour (sports gear + road signs), most weapons
- **Sci-Fi Elite?** Pristine minimalist suit, holographic accessories, perfect condition
- **Background Survivor?** Patched, weathered, makeshift repairs



Adam Smasher, cyberpunk corporate boss. *Cyberpunk 2077*

When designing a game boss you can enhance these markers of hierarchy. For example this cyberpunk corporate boss, he has the most expensive suit, integrated cyberware and genuinely looks strong and imposing.

For other characters, like for every period so far, think about their importance in the game's story and play into that.

A street merchant can be dressed looking like whatever he specialises in, but not over the top, he's just a merchant.

A raider leader would need to show he's the leader so the best armour, the most weapons etcetera.

Game examples

- **Cyberpunk 2077:**
 - Corporate district (sleek, clean)
 - vs. Watson district (grungy, DIY)
- **The Last of Us:**
 - WLF (organized, matching gear)
 - vs. Seraphites (makeshift, tribal)
- **Destiny 2:**
 - Exotic gear (glowing, detailed)
 - vs. common gear (simple, plain)

WLF >



Seraphites >

One of the game examples I thought was done well was the difference in factions in the last of us.

Here are pictures from the tv series but in the game I thought it was also done well.

The wlf is an organised group of rebel militia, wears matching gear, weapons, and show they have control in their outfits

While the seraphites, a brutal, fanatical religious cult, dresses in foraged clothing, more tribal but practical for their lives in survival mode

Question

- Can you think of another example in media (games, series, movies) where the characters are clearly recognisable in terms of hierarchy?



Star Wars Andor



Materials, textiles & dyes

Section 4

My next topic is materials, textiles and dyes

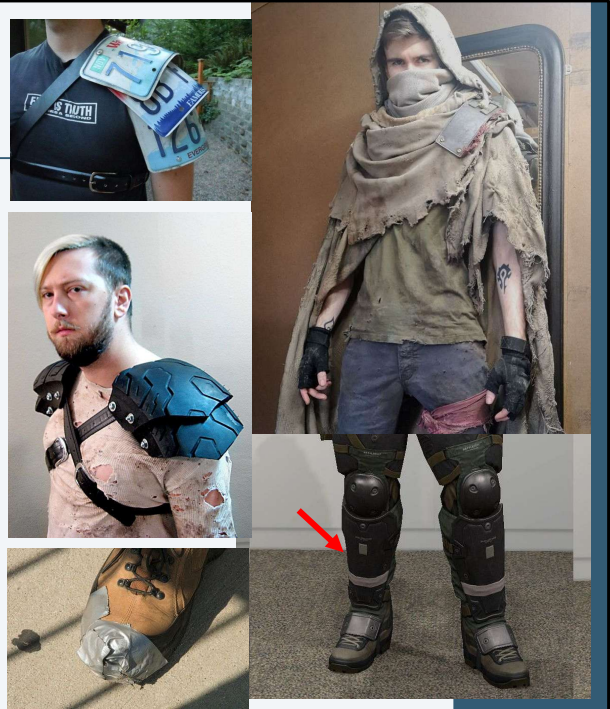
Materials POST-APOCALYPTIC:

Scavenged Material Logic:

- **Pre-War Materials** (Still exists but weathered)
- Denim, leather, cotton - 50+ years old
- Faded, patched, heavily worn

Repurposed Materials:

- **Tire rubber** - Armor plates, soles
- **Sports equipment** - Shoulder pads, shin guards
- **Road signs** - Metal plating
- **Plastic bottles/caps** - Cut into scales
- **Duct tape** - Holds EVERYTHING together



In a post apocalyptic world, pre-war materials still exist but are weathered strongly. All natural fibers would be degrading and faded. Need to be patched to be practical.

Some key repurposed materials in the world would be tires, sports equipment, road signs, parts of bottles and duct tape.
These can be diy'd into armour or things you need to simply survive

Materials

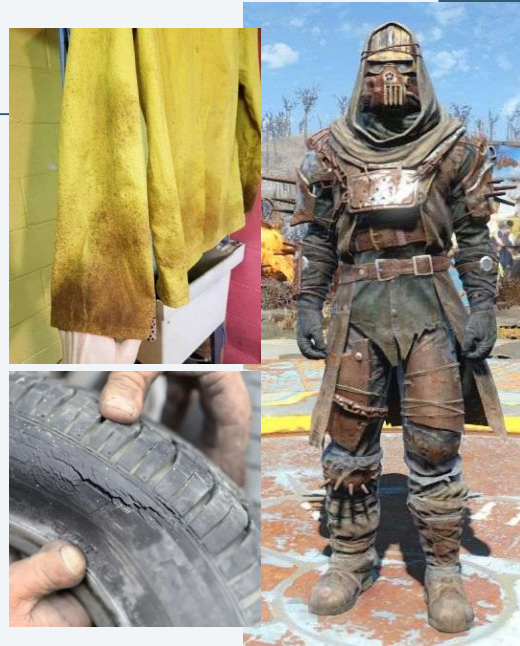
POST-APOCALYPTIC:

Makeshift Materials:

- Scrap metal, mesh, chains
- Whatever survived

Material Behaviour:

- Dried out, brittle
- Sun-bleached
- Water-damaged in some areas
- Decades of dirt/grime



Other makeshift materials can be scrap metal, mesh and chains. Just whatever still survives when nothing new is produced anymore

The material behaviours that you would see are dried out materials, like rubber or leather. Fabrics can be sun bleached or water damaged.

Often what people wear will show the decades of dirt and grime build up as there is no time or place to clean things often enough.

Materials

CYBERPUNK (Near Future)

Speculative Materials:

Smart Fabrics

- Color-changing (thermochromic, electrochromic)
- Reactive to environment

Ballistic Weave

- Soft fabric with protective properties
- Hidden armour

Synthetic Leather

- Improved beyond modern leather
- Can mimic perfectly or look intentionally synthetic



Then going into the near future cyberpunk era, here the speculative materials begin

Think of smart fabrics that can change colour or react to the environment

Different weaves that give the fabrics protective properties of hide things

Or synthetic leather which can become anything you need it to be while still following the behaviour of modern leather

Materials

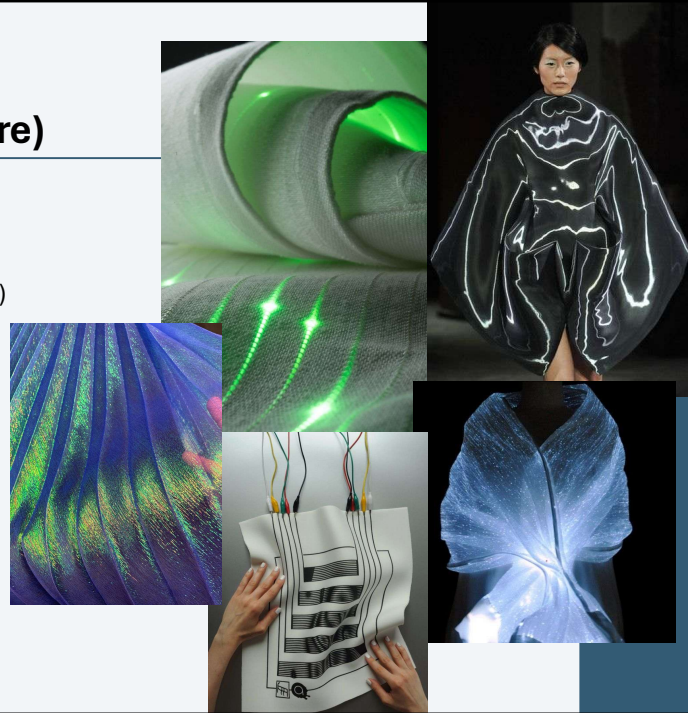
CYBERPUNK (Near Future)

Tech Integration Materials:

- Conductive fabrics (visible circuits)
- Fiber optic weaving (glowing threads)

Holographic Materials

- Iridescent, colour-shifting
- Status symbol (expensive)



You will start to see tech integrated materials, like visible circuits or glowing threads.

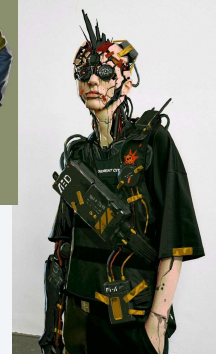
And holographic materials, that have so many different colours and could be a status symbol

Materials

CYBERPUNK (Near Future)

Material Aesthetic by Status:

- **Corporate Elite:** "Natural" (cotton, wool) = EXPENSIVE (post-scarcity makes natural rare)
- **Street Level:** Synthetic everything (cheap, practical)
- **Tech-focused:** Obvious tech integration, function over form



Cyberpunk, like most others, status can be shown through material aesthetic. Natural fabrics would be very expensive, often seen in very minimalistic outfits just for aesthetics

Synthetics are still very cheap but practical

And obvious tech integration prioritizes function over form.

Materials SCI-FI FUTURE (2100+):

Advanced Speculative Materials: Nano-Weave

- Molecular-level construction
- Self-repairing, adaptive
- Can be bulletproof while feeling like silk

Energy Field "Fabrics"

- Hard-light constructs
- Can appear/disappear

Bio-Engineered Materials

- Grown, not woven
- Living fabrics (self-cleaning, temperature-regulating)



Then the sci fi future materials, this is far into the future
You could think of nano weave materials, which could be self repairing,
adapative, bullet proof while still mimicking the fluidity of silks
There could be energy field fabrics made of hard light, which can appear and
disappear like a lightsaber
Or bio engineered materials that are grown in a lab. These could be alive and
caring for you or the fabric

Materials

SCI-FI FUTURE (2100+):

Memory Materials

- Return to shape
- Wrinkle-proof taken to extreme

Programmable Matter

- Can change form/texture on command

Printed materials

- Supporting shapes
- Open yet strong



Memory materials that return to shape, you can take wrinkle proof to the extreme
Programmable matter is cool, this can change form or texture on demand. Might be useful for a character that needs to shapeshift or be super stealthy, hiding in plain view

And printed materials, can be open yet very strong.

Fabrics

Synthetic Materials Explosion

Nylon (1930s)

- First fully synthetic fiber
- Strong, elastic, water-resistant
- Stockings, parachutes, eventually everything

Polyester (1950s)

- Wrinkle-resistant, durable, cheap
- Can mimic other fabrics

Spandex/Lycra (1958)

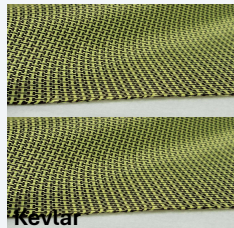
- Stretch, form-fitting
- Athletic wear, superhero suits

Gore-Tex (1970s)

- Waterproof + breathable
- Technical outdoor gear

Kevlar (1960s)

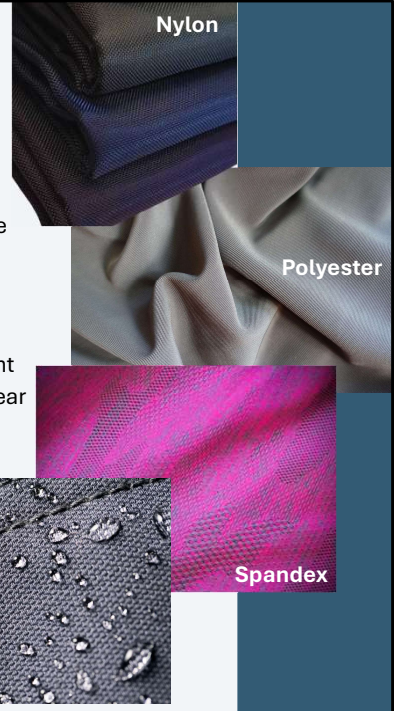
- Bulletproof, cut-resistant
- Body armour, tactical gear



Kevlar



Gore-Tex



I just talked about materials and now I'll tell you about the fabrics

In the last 100 years the synthetic materials exploded. A lot of new stuff was invented which we still use today

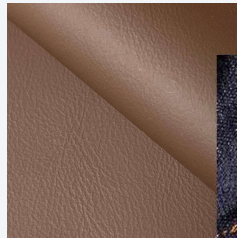
These are designed to offer a certain thing, like protection, strength, water repellancy, form fitting stretchiness etcetera

These are the materials you can use as the base when you're designing something entirely new.

Fabrics

Modern Natural Materials (Still used):

- Cotton, wool, leather - improved processing
- Denim - workwear to fashion staple
- Technical leather (treated for performance)



Modern natural materials will still be used but will have improved the process, like sustainable farming, stronger materials, or a shift in purpose as you see with denim which went from only workwear to a fashion staple everyone wears today

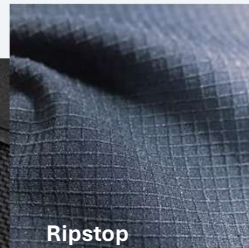
Fabrics

TACTICAL/MILITARY MATERIALS:

- **Cordura** - Heavy-duty nylon, abrasion-resistant
- **Ripstop** - Grid reinforcement pattern (visible)
- **Ballistic nylon** - Dense weave, protective
- **Molle webbing** - Modular attachment system



Cordura



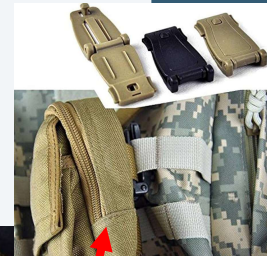
Ripstop



Ballistic nylon



Molle webbing

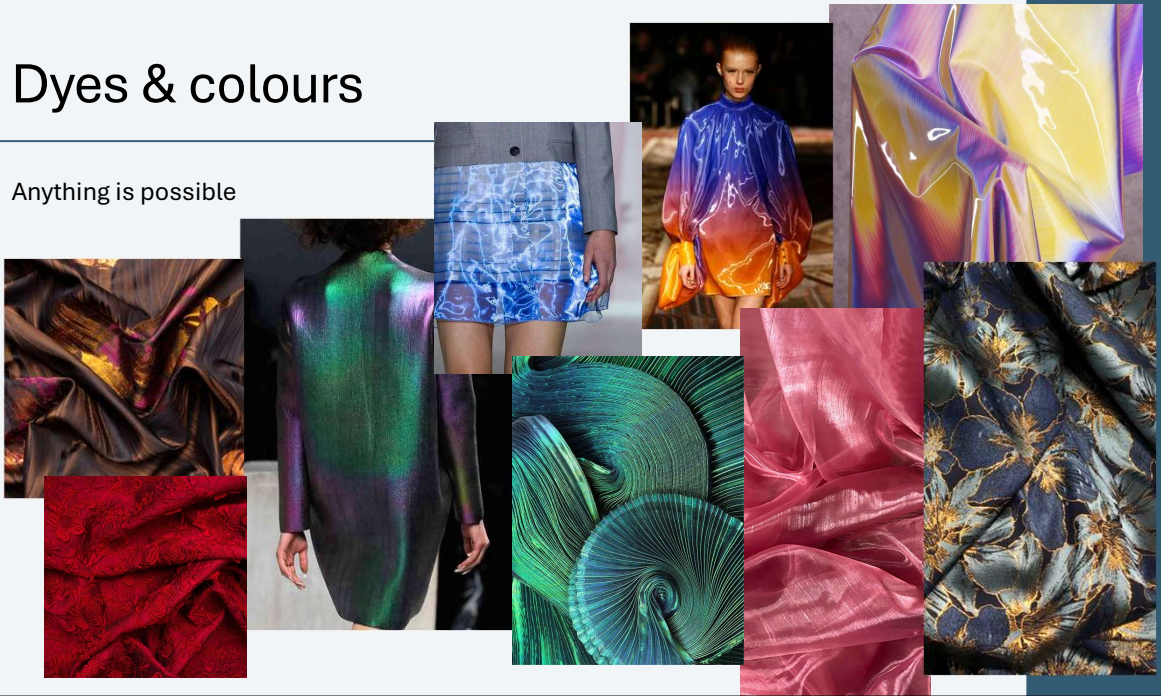


Then onto the tactical and military fabrics.

Here you can find some more interesting fabrics to take as a base like Cordura, ripstop, ballistic nylon or mole webbing (which is more an system than a fabric) Molle webbing is a modular attachment system where you can attach bags, gear, equipment, weapons to a structure using Velcro, straps or clips and buckles You can design this in many different shapes or forms, try something different from just straps sewing in lines.

Dyes & colours

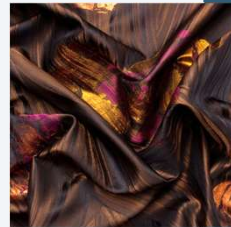
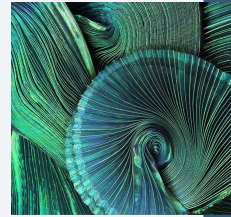
Anything is possible

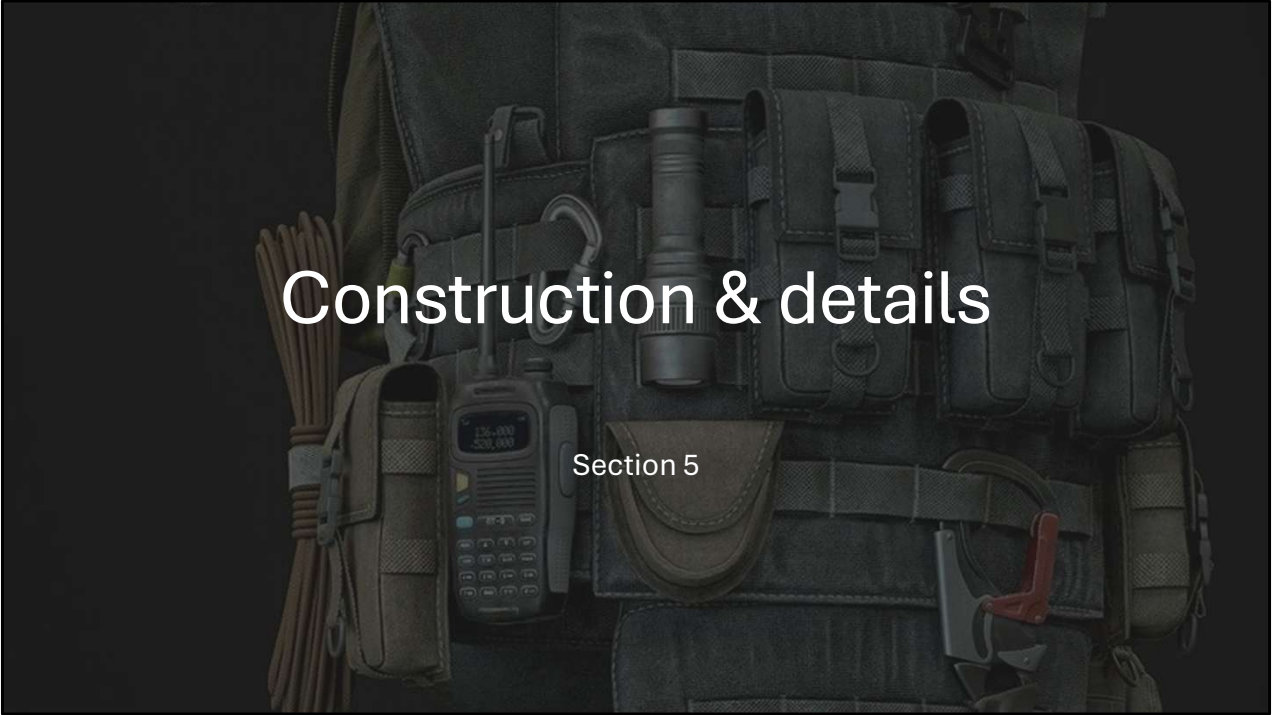


In terms of dyes and colours, anything is possible. Be imaginative, what new technology could influence this in your world?

Question for you

- How would you apply and use these concepts to what you're currently working on?
- Or how is this useful within your discipline?



A tactical vest with various equipment including a radio, flashlight, and tools.

Construction & details

Section 5

Construction & details

Modern (1900s-present):

INDUSTRIAL TAILORING:

Key Innovations:

Sewing Machines (Uniform Stitching)

- Perfectly straight seams
- Consistent tension



Zips (1930s onward)

- Quick closure
- Centre front or side placement
- **Animation:** Rigid line (doesn't stretch)
- Can catch/jam (detail for wear)

Velcro (1950s)

- Quick release
- Adjustable
- **Animation:** Peeling effect, strips can separate



Elastic

- Waistbands, cuffs, gathers
- Stretch and return
- **Animation:** Compression around body, stretches with movement



"haberdasheries"

Modern times invented the details we know well today like Velcro zippers and elastic.

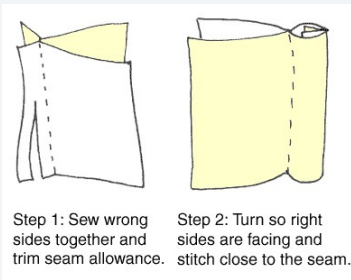
I've added some information that might be useful when designing, sculpting or researching these.

---These are called haberdasheries, which is kind all the hardware of a garment, so that's a term you can google to find more of this if you're not specifically sure what you want to use yet.

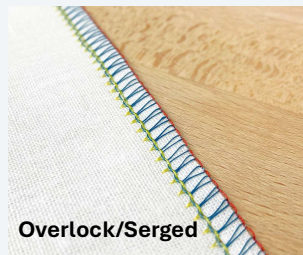
Construction & details Modern (1900s-present):

MODERN SEAM TYPES:

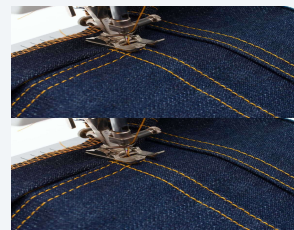
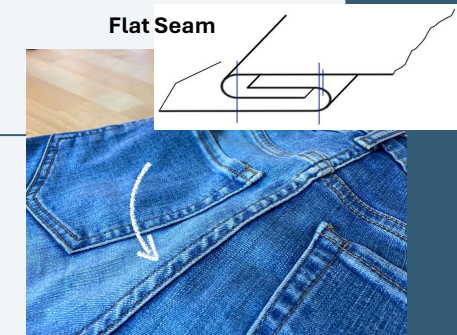
- **French Seam** (clean finish)
- **Overlock/Serged** (prevents fraying)
- **Flat Seam** (no bulk, athletic wear, super strong)
- **Topstitching** (decorative + reinforcement)



French Seam



Overlock/Serged



Topstitching

Some seam types that can be seen in the modern period are French seams, this gives a clean finish. Often done with sheer dresses or other garments where overlocking is not pretty enough

Then overlocking, this is the edge you get in most other garments. Mostly knits.

So if you're wearing a t-shirt or hoodie right now, check the seam finish

Then there is the flat seam, this is an interesting one and because its construction is so many layers, this one is super strong. If you're wearing jeans, it's the seam that runs between each side of the pant legs. In the seat

And then there's topstitching. This can be decorative and offers reinforcement to the seams. You can choose a matching colour or deliberately choose a contrasting colour

Construction & details

Modern athletic/performance:

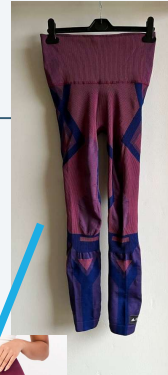
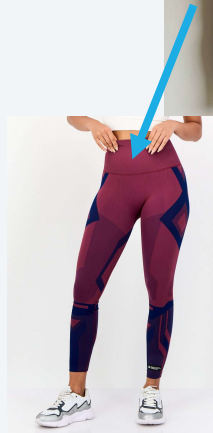
Key Elements:



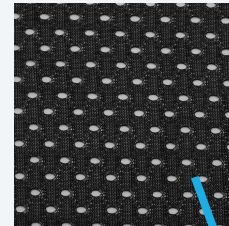
< **Panel Construction**
Multiple small pieces



^ **Stretch Direction**
Four-way stretch



< **Compression Areas**
Tight fit specific zones



Mesh Panels >
Ventilation

Then here are the key elements of athletic and performance wear

Panel constructions are popular, offering better fit.

These garments often use a 4 way stretch

Or compression areas, this makes specific areas tighter and offers support. As you can see in the red pants picture, it looks quite funny when not worn

And mesh panels are quite popular for extra ventilation

Construction & details

Tactical/military

PLATE CARRIER CONSTRUCTION:

Layers:

1. Base vest (fabric)
2. Armour pockets (rigid plates slide in)
3. MOLLE webbing (external)
4. Pouches (attached)



< Plate

The plate carrier construction is quite interesting I think

This is a base vest which has armour pockets for the plates, hence the name plate carrier

And on the outside has molle webbing so you can attach all the tools, ammo, gadgets you need when dispatched

Because this vest is so rigid and quite heavy due to the plates, movement is somewhat limited in the upper body and soldiers need to keep a straight posture . The vest is often cropped so from below the ribs the movement is still flexible.

Construction & details

Tactical/military

KEY DETAILS:

- Velcro patches (rank, name)
- D-rings/carabiners (attachment points)
- Adjustable straps (shoulders, sides)
- Drawstrings (quick adjustments)
- Press studs (popper buttons)



Some key details in military clothing are Velcro patches, d rings or carabiners, adjustable straps, drawstrings and press studs

Construction & details Post-apocalypse

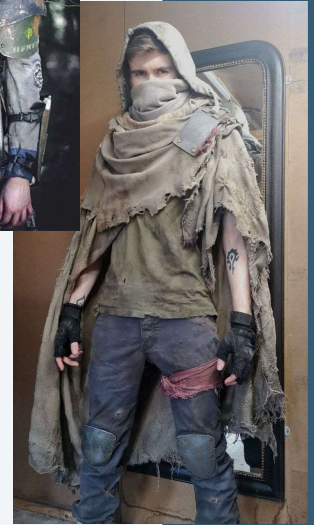
SCAVENGED ASSEMBLY:

Construction Logic:

- No manufacturing = improvisation
- Whatever fits, works
- Layering for protection

Key Elements:

- Improvised Armour
- Visible Repairs
- Layering for Bulk
- Exposed Fastening
- Weathered, teared, fraying, holes



In the post apocalypse, there is no more manufacturing so everything needs to be improvised. Whatever fits works.

If it is layered for protection and scavenged clothing often doesn't give enough protection in itself so layering it gives both benefits

The key elements are improvised armour, visible repairs, layering for bulk, exposed fastenings and a weathered look with tears, fraying or holes in it.

Construction & details Cyberpunk (near future)

CYBERPUNK CONSTRUCTION:

Layering philosophy:

1. Base layer (tight, technical)
2. Mid layer (functional, pockets)
3. Optional: Outer layer (asymmetric, style)

Key construction elements:

- Asymmetric Closures
- Modular Attachments
- Tech Integration Points
- Mixed Materials
- Cybernetic integration



The cyberpunk layering philosophy that can often be observed is a tight base layer, then a functional mid layer with pockets and an optional outer layer, like a jacket for style and contrast

Key construction elements could be Asymmetric Closures
Modular Attachments Tech Integration Points Mixed Materials And Cybernetic integration

Construction & details Sci-fi future (2100+)

SEAMLESS CONSTRUCTION:

Advanced Techniques: 3D Printing/Grown Garments

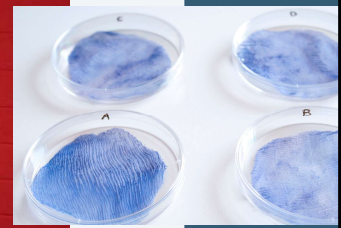
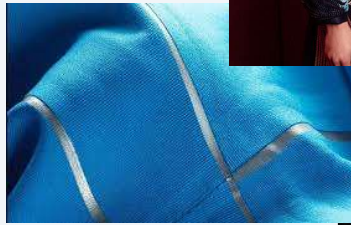
- No seams (one piece)
- Integrated structure

Bonded Construction

- Pieces fused, not sewn
- Creates clean lines

Modular Hard Armour

- Plates clip onto bodysuit
- Magnetic or mechanical attachment



^ Bio grown fabric
< 3D printed fabric



Then the sci fi future

This era has seamless construction like 3d printing or grown garments which are one piece or integrated structures

There could be bonded constructions, the pieces are fused and not sewn

And modular hard armour that can clip onto each other or the garment underneath with magnetics or with something mechanical

Construction & details

Sci-fi future (2100+)

Smart fabric construction

- Programmable shape
- Active fit adjustment

Bodysuit foundation:

- Form-fitting base layer
- Minimal seams (under arms, sides)
- Armour/tech attaches over

Hard-light constructions:

- Not physical fabric
- Energy field



Smart fabrics that can be programmable in shape for an active fit adjustment
You can think of the bodysuit as a foundational piece, that every other thing attaches to
Or use hard light constructions which are not physical fabrics, like light sabers, imagine what a garment made from light saber energy would look like or could offer

Designing for practicality

- **Modular systems** - swap gear/tech without replacing entire garment (MOLLE, magnetic clips, ports)
- **Multi-functional materials** - one fabric serves multiple purposes (bulletproof looks normal, smart fabrics regulate temp + display info)
- **Reinforced stress points** - double-layer at knees, elbows, palms for durability during movement
- **Quick-release closures** - rapid equipment changes or emergency removal (velcro, magnetic, buckles)
- **Integrated tech channels** - cables, ports, charging built into construction (prevents tangling/damage)
- **Repairable/replaceable** - modular panels swap when damaged, or self-repairing materials

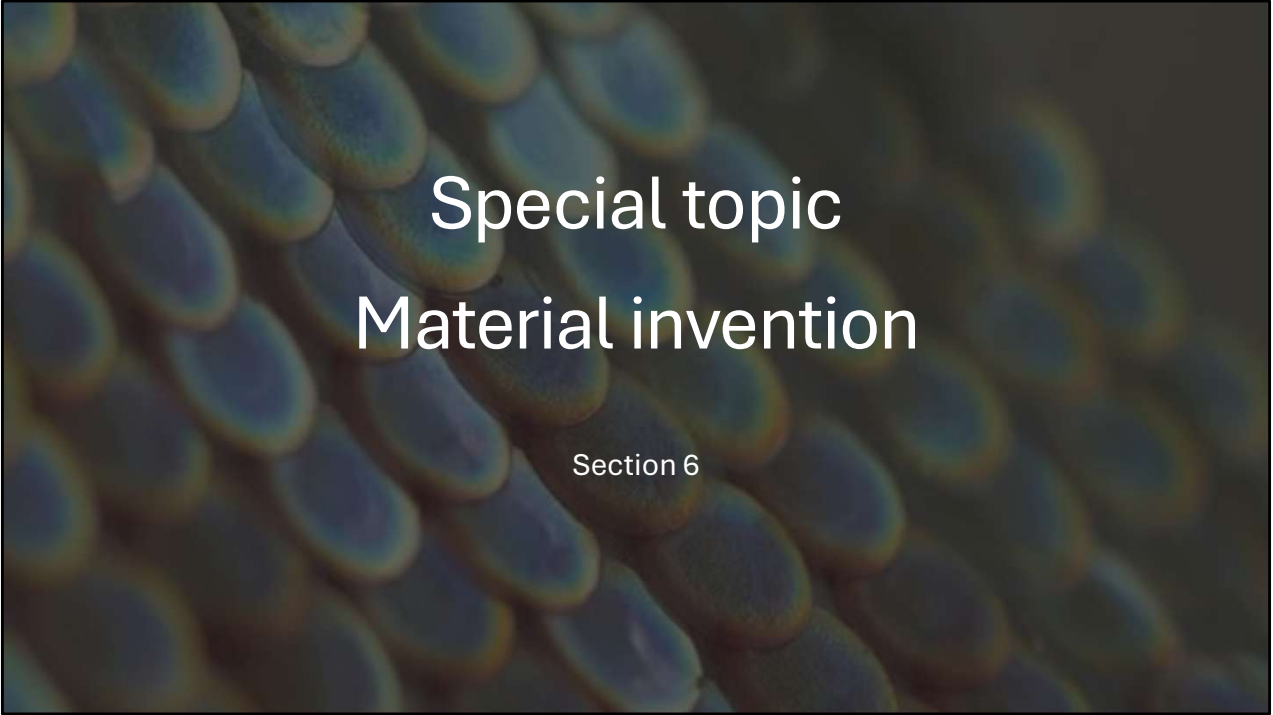


When designing for this era, practicality is very important. As I told you just now, there are a bunch of ways to make a garment more practical like using modular systems, multi functional materials, you can reinforce stress points for durability, quick release closures for rapid equipment changes. Tech can be integrated visible like you see in the image or the garment or armour is fully modular.



Reflection prompt

What did you learn that was new and really cool that sparked curiosity and new ideas?



Special topic

Material invention

Section 6

The special topic of today is material invention. How do you design a material that has no grounding in real life yet?

Special topic introduction

Why invent new materials?

When You Need Speculative Materials:

- Far future settings (no Earth historical reference)
- Alien cultures (what would THEY develop?)
- Post-apocalyptic worlds (scavenged + invented combinations)
- Advanced technology settings (materials that don't exist yet)
- Fantasy with unique magic systems (enchanted/living materials)

The Design Challenge: How do you make something that doesn't exist feel believable and grounded?



So why should you invent new materials?

You might need them when designing a far future setting which has no earthly reference. Or alien cultures that are completely different from humans.

A post apocalyptic world where everything is scavenged and invented through diy;ing

It could be useful when designing for advanced technology settings, like the industrialisation, what would suddenly be possible with such a new shift in tech? Or a fantasy setting where unique magic systems enhance the possibilities in fabrics and construction possible

This gives you a design challenge. How do you make something that doesn't exist feel believable and grounded?

Step 1: identify the purpose

Ask: What problem does this material solve in your world?

Examples:

- Cyberpunk street merc needs: Bulletproof protection + mobility + affordable
- Space explorer needs: Temperature regulation + radiation shielding + lightweight
- Post-apocalyptic survivor needs: Protection + durability + repairable with scavenged materials
- Sci-fi soldier needs: Ballistic protection + environmental adaptation + tech integration

Define the functional requirements:

- What must it protect against? (Bullets, blades, environment, energy weapons)
- What conditions must it survive? (Heat, cold, vacuum, water, radiation)
- Who can afford/access it? (Elite military, common citizen, scavenger)
- What are the trade-offs? (Heavy vs. light, protective vs. mobile, expensive vs. accessible)

Game Application: This is your design brief - purpose drives every other decision



First you need to identify the purpose. What problem does the material solve?
Here you can read some examples

And you need to define the functional requirement from the problem it needs to solve

What does it protect against? What conditions must it survive? Who can afford or access it? What are the trade offs?

You can see this as the design brief. Every purpose drives your decisions

Step 2: choose real-life base references

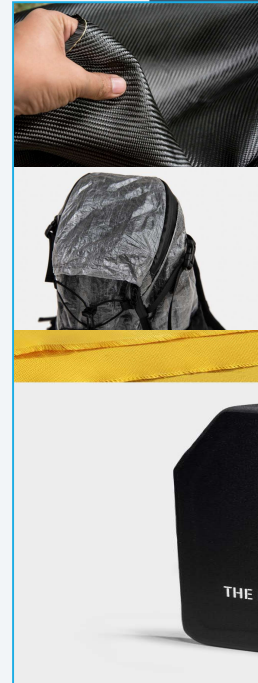
Find existing materials with similar properties:

For Protective Materials:

- **Kevlar** - Bulletproof, cut-resistant, woven aramid fibers (stiff, structured)
- **Dyneema** - Ultra-strong, lightweight, used in body armour (flexible but protective)
- **Nomex** - Fire-resistant, used in firefighter/race suits (protective against heat)
- **Ceramic plates** - Hard armour inserts (rigid, heavy, cracks on impact)

For Performance Materials:

- **Gore-Tex** - Waterproof + breathable (membrane technology)
- **Neoprene** - Insulating, flexible (wetsuit material)
- **Compression fabrics** - Lycra/spandex blends (support, enhance performance)
- **Merino wool** - Temperature regulating, moisture-wicking (natural high-performance)



Then, you need to find real life materials that can be used as the base

Find materials with similar properties

If you need something protective choose from Kevlar, dyneema, nomex or ceramic plates.

If you need something performative, choose gore tex, neoprene, compression fabrics or even a natural fabric like merino wool.

Step 2: choose real-life base references

For Adaptive Materials:

- **Shape-memory alloys** - Return to original shape when heated
- **Thermochromic materials** - Change colour with temperature
- **Phase-change materials** - Absorb/release heat to regulate temperature

Why Use Real References:

- Gives you vocabulary (woven, membrane, composite, layered)
- Provides visual/textural starting point
- Grounds the invention in physics (how it might actually behave)
- Players recognize the "familiar" elements even in futuristic context

Game Application: "This feels like it could exist" comes from basing on reality



If you need an adaptive material, you can think of shape memory alloys, thermochromic materials or phase changing materials

You should use real references because it gives you the vocabulary you can build on. It provides a visual and textural starting point and grounds the invention in physics.

Players will recognise the familiar elements which gives you easier believability. So making something that feels like it could exist, comes from basing it on reality.

Step 3: extrapolate into your world

Take real material properties + push them further:

Example 1: Bulletproof Street Wear (Cyberpunk)

- **Purpose:** Protection for street mercs, affordable, everyday wear
- **Real base:** Kevlar (bulletproof) + athletic compression wear (flexible)
- **Future extrapolation:**
 - Woven at nano-scale (thinner, more flexible than current Kevlar)
 - Could look like normal clothing (hoodie, jacket) but protective
 - Still has limitations: Stops bullets but impact still hurts (bruising), expensive for street level (mid-tier protection), shows wear/damage over time
 - **Visual result:** Looks like modern tactical wear but slightly too perfect/smooth texture, maybe subtle woven pattern visible up close



Then step 3, take the real materials and push them further
Here the example of cyberpunk bullet proof street wear
(read the slide)

To me the texture of this looks exactly like a phone battery, the blackness the shine of it, the half shiny half matte quality. Maybe it offers power too, who knows

Step 3: extrapolate into your world

Example 2: Self-Regulating Environmental Suit (Space/Sci-Fi)

- **Purpose:** Survive extreme temperatures, breathable, comfortable for extended wear
- **Real base:** Gore-Tex (waterproof/breathable) + phase-change materials (temperature regulation)
- **Future extrapolation:**
 - Active temperature control (not just passive regulation)
 - Self-sealing if punctured (like modern self-healing polymers but advanced)
 - Bio-monitoring integrated (fabric reads vitals)
 - Trade-off: Requires power source (battery pack visible), expensive (elite only), complex (can malfunction)
 - **Visual result:** Sleek bodysuit with subtle tech patterns, possible glowing elements at power points, seamless construction



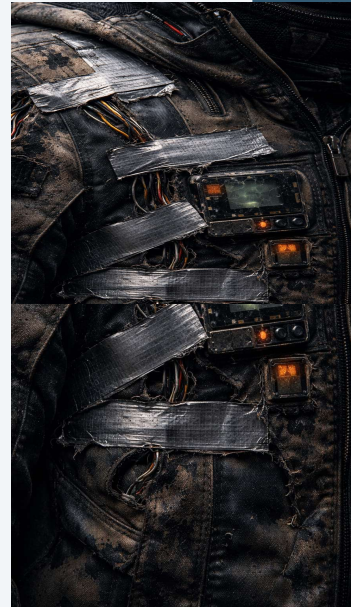
Then another example, a self regulating environmental suit

[read slide]

Step 3: extrapolate into your world

Example 3: Scavenged Smart Fabric (Post-Apocalyptic)

- **Purpose:** Survivors found old military/corporate tech fabric, repurposing it
- **Real base:** Modern technical fabrics (Gore-Tex, Cordura) aged 50+ years
- **Future extrapolation:**
 - Was adaptive/colour-changing when new, now broken/malfunctioning
 - Shows age: Faded, tears, some tech elements dead (no power), patched with non-tech fabric
 - Still somewhat functional (water-resistant coating remains, stronger than regular fabric)
 - Trade-off: Can't repair properly (no manufacturing), limited quantity, uneven deterioration
 - **Visual result:** High-tech fabric with visible damage, colour inconsistent, mixed with crude repairs, duct tape holding broken tech elements



And the final example scavenged smart fabrics
[read slide]

Making it visually believable

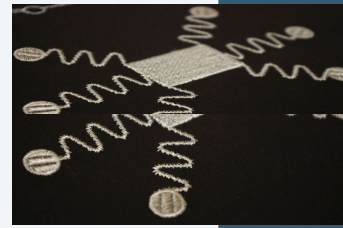
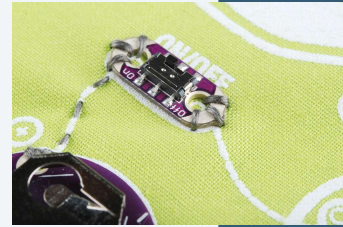
Visual Design Principles for Speculative Materials:

1. Show the Technology Integration

- Where does power come from? (Visible battery packs, charging ports, energy cells)
- How does it connect? (Seams with conductive threading, plug-in points, magnetic attachments)
- What happens when it breaks? (Glitching patterns, dead zones, visible damage)

2. Maintain Material Logic

- Heavy materials drape with weight (even if futuristic)
- Rigid materials don't bend (unless that's the innovation)
- Flexible materials show stress/stretch points
- Everything has texture (perfectly smooth = deliberately shows advanced tech)



A few things you can use to make it visually believable when designing speculative materials

Show the tech integration, if the garment has power, show where it comes from. Show how it connects. Think of what happens when it breaks

Try and maintain the material logic, remember those sliders I talked about?

Weight, drape, bending, stretch, shearing, all still apply to speculative materials
And think of the texture, smooth might indicate advanced tech and roughness something older, more weathered

Making it visually believable

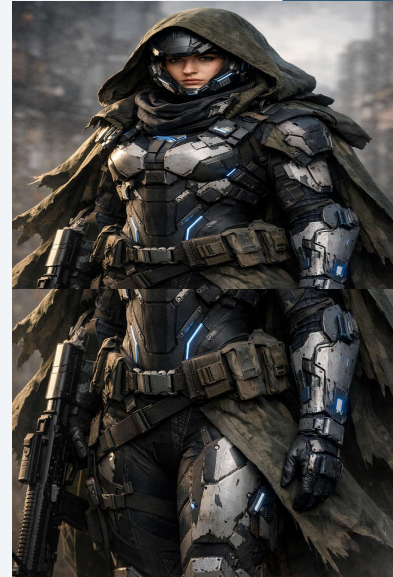
Visual Design Principles for Speculative Materials:

3. Design the Limitations

- Perfect materials are boring and unbelievable
- What's the trade-off? (Protective but heavy, adaptive but expensive, powerful but requires maintenance)
- How does it fail? (Tears, power loss, degradation over time)
- Who CAN'T access it? (Creates visual hierarchy - not everyone has best tech)

4. Consider Layering

- Futuristic doesn't mean single-layer
- Base layer (bodysuit, tech integration) + protective layer (armour) + environmental layer (jacket, cloak)
- Each layer can be different material with different properties



Find the limitations, perfect materials are boring what what is the trade off? Does it ever fail and how? Who cant access it? Not everyone should be able to afford everything. Think of the hierarchy

And consider layering, even in the future, an outfit is still not a single layer. Each layer can have different uses, properties and materials to tell its story

Game examples breakdown

Example: Cyberpunk 2077 - Netrunner Suit

- **Purpose:** Tech interface + protection + shows status
- **Real base:** Athletic compression wear + ballistic weave
- **Speculative:** Circuit patterns (connectivity), subtle glow (active tech), too-perfect surface (nano-weave)
- **Limitations:** Expensive (street level can't afford), requires maintenance, visible damage shows use

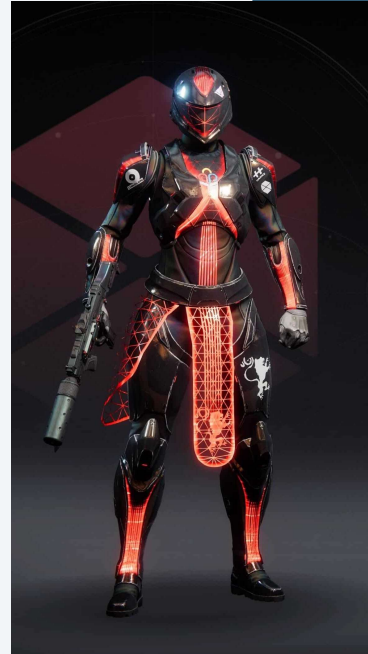


Here is another example. The cyberpunk netrunner suit
You can read its purpose, the tech, protection and status
It's designed from the real base of athletics
Its speculative component is the circuits and too perfect surface
And the limitations are that it's super expensive, requires maintenance and
damage shows clearly

Game examples breakdown

Example: Destiny 2 - Guardian Armour

- **Purpose:** Protect against alien weaponry + show class identity + integrate Light powers
- **Real base:** Tactical armour + motorcycle protective gear + medieval plate logic
- **Speculative:** Self-repairing (lore), energy-reactive (glows with abilities), modular (swap pieces)
- **Limitations:** Different quality tiers (exotic vs. common visually distinct)



And here another example, this one is from destiny.
The purpose shows with class identity
Its based on the real base of tacticle armour and motorcycle gear
The speculative part is grounded in lore, shows abilities and can be modular
And the limitations are the quality tiers

Your design workflow

When Designing a Speculative Material:

Quick Reference Method:

1. **Define the need** (1 sentence: "This character needs X because Y")
2. **Find 2-3 real materials** that have similar properties (name them, show reference images)
3. **List what you're keeping** from real materials (flexibility of lycra, protective quality of Kevlar, temperature regulation of wool)
4. **List what you're advancing** (nano-scale weaving, self-repair, colour-changing, power integration)
5. **Define the limitation** (What's the cost? What fails? Who can't access it?)
6. **Sketch/describe visual result** (Texture? Sheen? Colour? Patterns? Tech elements?)

A quick recap of your possible design workflow when designing speculative materials

1 define the need in a sentence

Find a few real materials that have similar properties, with names and images

List what you're keeping like flexibility or temperature regulation

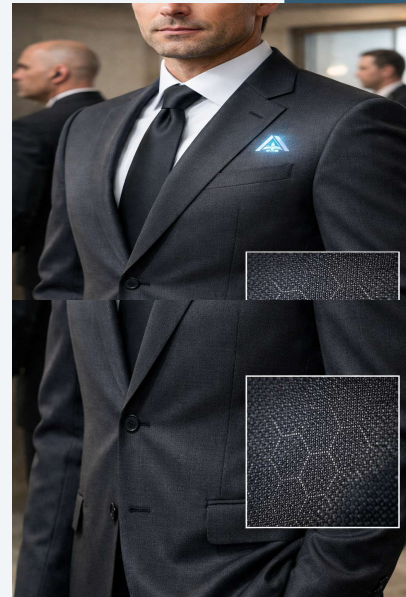
List what you're advancing, the weaving, self repair properties of power integration

Then define the limitation and lastly sketch or describe the visual result.

Your design workflow

Example Workflow Applied:

1. **Design brief:** "Elite corporate security needs discreet protection in business setting"
2. **Real bases:** Kevlar (protection), merino wool (looks normal), tailored suiting (business appropriate)
3. **Keeping:** Woven structure, professional appearance, flexibility
4. **Advancing:** Nano-weave bulletproof in suit-weight fabric, adaptive fit, subtle tech monitoring
5. **Limitation:** Extremely expensive (executive class only), requires corporate tech infrastructure, can be hacked/disabled
6. **Visual result:** Looks like expensive business suit, too-perfect drape, very subtle tech pattern visible up close, possible holographic corporate logo, impeccable condition (self-cleaning)



Here the workflow in an example.
[read slide]

Helping questions:

1. What problem does this material solve in this specific world?

Protection? Temperature regulation? Status display? Resource scarcity? Environmental hazard?

2. What are the trade-offs?

If it's protective, is it heavy? If it's adaptive, does it need power? If it's advanced, is it expensive/rare? Nothing is perfect - what does this material sacrifice to gain its primary function?

3. What would THIS species/culture actually invent based on their biology?

What would a lizard-person invent for fabric? They're cold-blooded - do they need heat-retaining materials? Their skin sheds - does fabric need to accommodate that? They have scales - would they even wear fabric or develop scale-enhancement technology instead?

4. What existing real-world materials have similar properties to what you need?

Find 2-3 real materials to reference - this grounds your invention in physics and gives you visual/textural vocabulary

5. How does this material behave physically?

Does it drape? Is it stiff? Does it stretch? How does it fold? How does light interact with it? Even invented materials need consistent physics.

6. Who has access to this material, and what does a LOWER-TIER version look like?

If this is elite-level material, what do common people wear? Older version of the tech? Cheaper substitute? Knockoff with visible flaws? This creates visual hierarchy and world depth.



Here are some question that might help you design these speculative materials

1. What problem does this material solve in this specific world?

2. What are the trade-offs?

3. What would THIS species/culture actually invent based on their biology?

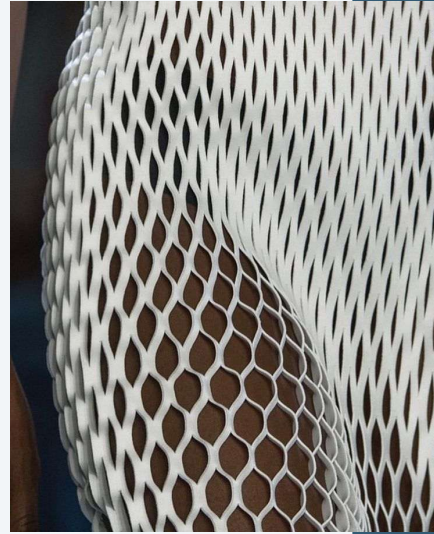
4. What existing real-world materials have similar properties to what you need?

5. How does this material behave physically?

6. Who has access to this material, and what does a LOWER-TIER version look like?

Question for you

- Was this guide on designing speculative materials useful and how will you use it in your work?



Discussion & reflection

Section 7



Reflection prompt

Does anything we covered today make you want to revisit something you've already made?



Review the Costume Design guild

1. Which guild was most useful for you personally?
2. And have you used anything in your work yet?
3. What topic did you find most surprising/unexpected?
4. (*hypothetically*) If I run this again next year, what would you change?
5. What worked really well and what didn't?



Thank you!!

- This was the last guild of block C
- Suggestions for block D?
- Can be:
 - More casual
 - Specific topics,
 - “work sessions”
 - Portfolio help
 - Or simply nothing at all



Thank you so much for being here and being part of my master thesis.
This was unfortunately the last guild of this block as next week is cancelled due to that international communication thing

Anyone suggestions for block D? Do you know what you'll be doing or what you need?

Could be more casual, specific topics, work sessions, portfolio help, or simply no guild at all.

Additional sources

How to set up a plate carrier – adjusted to personal movement preferences [\[link\]](#)

Write a design brief with real base, keeping, advancing, limitation and visual result description and feed it to ChatGPT to get design inspiration.